

Supplementary Material

Catalogue of Catalogues: A Web Platform for Earthquake Catalogue Management, Merging, and Statistical Analysis

This supplement collects additional screen captures of the *Catalogue of Catalogues* (CofC) platform that support, but are not essential to, the argument of the main paper. All captures are from release v0.1.0 running in a browser against a local instance.

Data. Every capture uses a reduced-scale synthetic catalogue pair (a 5,200-event “GeoNet-like” catalogue and a 4,300-event “Agency B” catalogue, sharing a 50% duplicate overlap): the same construction as the worked example of the main paper, at roughly one twenty-third of its scale so that it loads and analyses interactively in the browser. The seeded reproductions distributed with the source repository — the figure script and the two unit tests listed under Data and Resources in the main paper — build catalogues of other sizes, so they reproduce the workflow but not the specific event counts and estimates shown in these captures. No real event data are shown anywhere in this supplement.

Statistical analysis in the browser

Event-level quality index

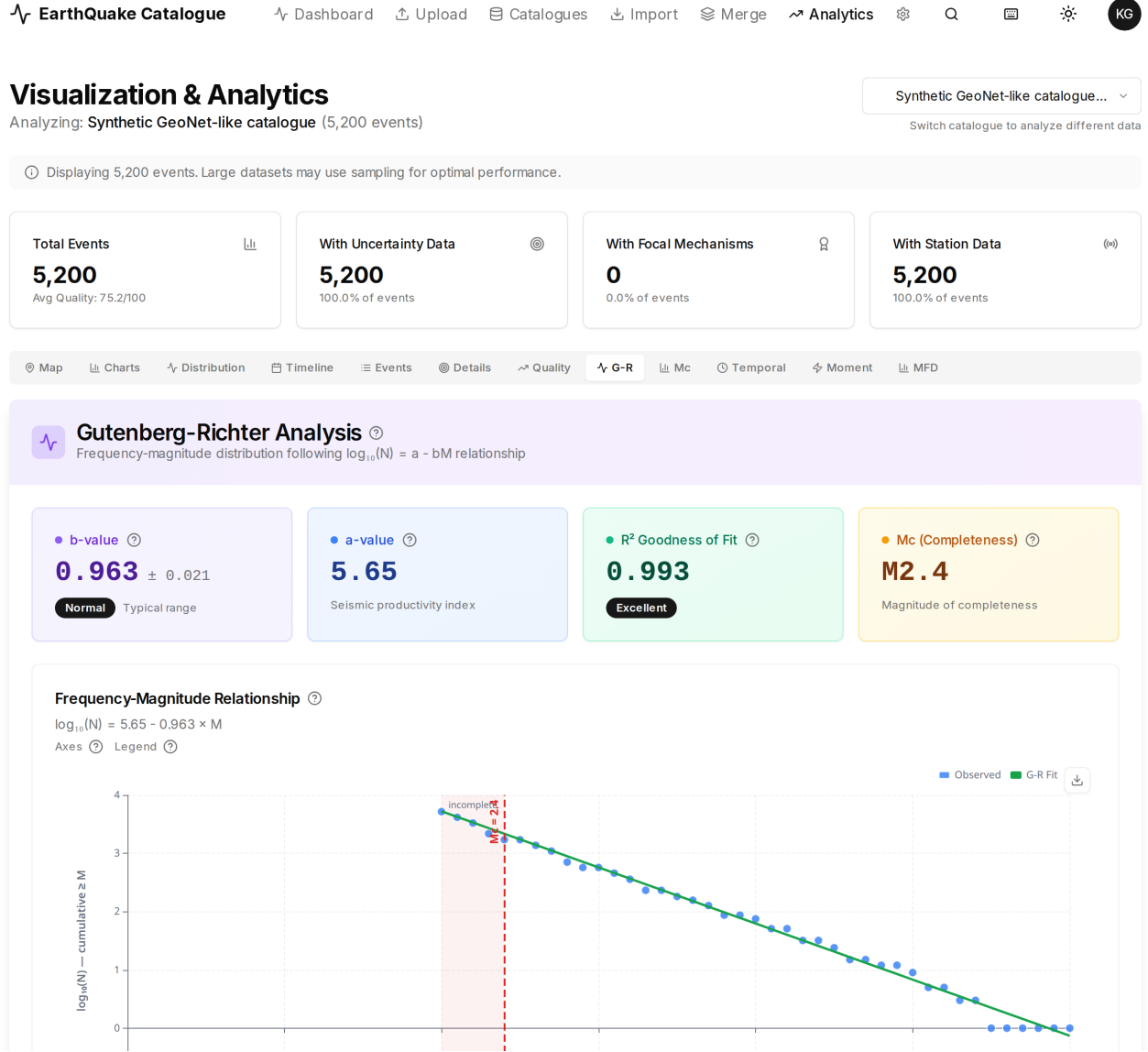


Figure S1: Gutenberg-Richter analysis of the synthetic GeoNet-like catalogue, computed client-side. The platform reports $\hat{b}$ with its counting-statistics uncertainty, the a -value, the goodness of fit, and the Maximum Curvature estimate of M_c ; the frequency-magnitude plot marks the incomplete range below M_c and overlays the fitted relation. The recovered $\hat{b} = 0.963 \pm 0.021$ is fitted above the platform’s own Maximum Curvature cut-off of $M_c = M2.4$ (Figure S2), on the 41.8% of the catalogue above that cut-off, and lies 1.8 counting-statistics standard errors below the $b = 1.0$ used to generate the synthetic catalogue. It is a check on the estimator rather than a measurement of New Zealand seismicity.

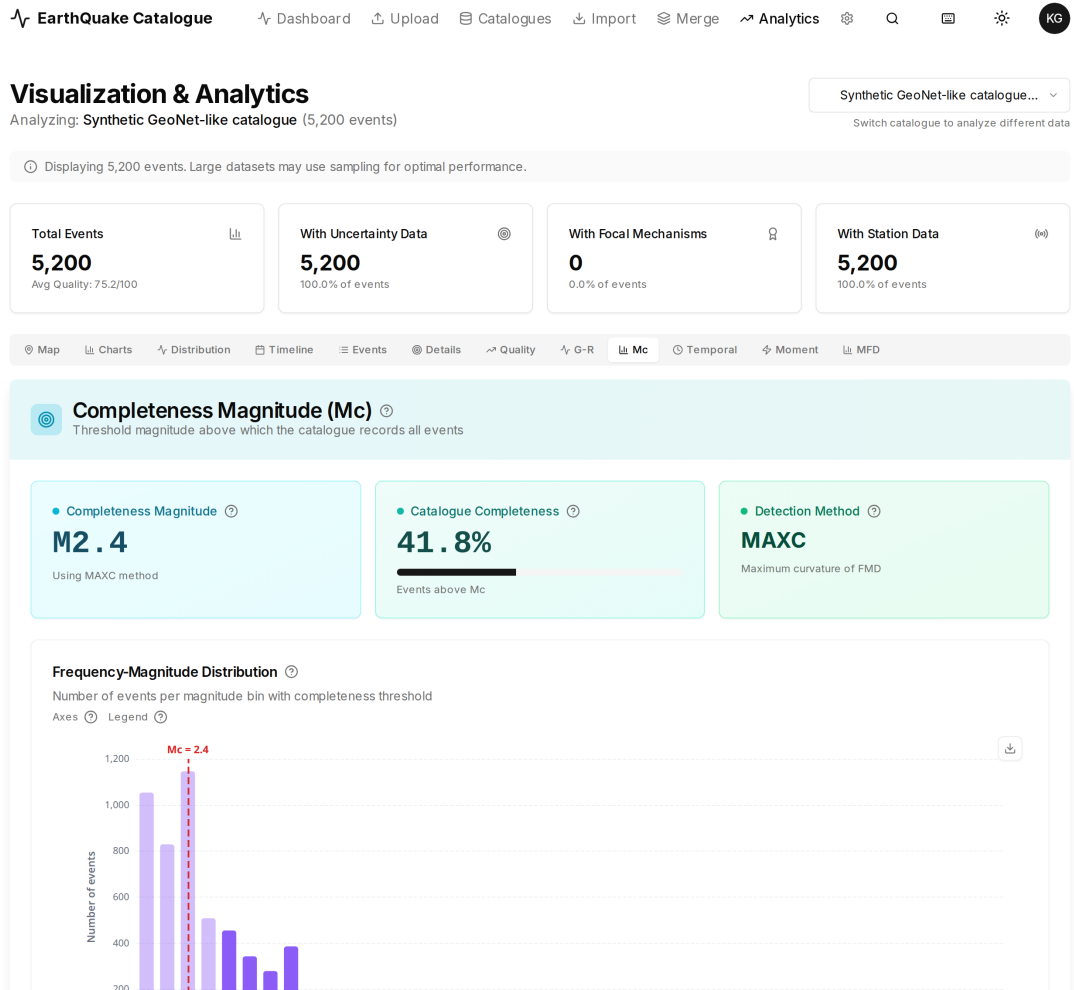


Figure S2: Magnitude-of-completeness panel for the same catalogue. M_c is estimated by Maximum Curvature with the +0.2 correction discussed in the main paper: here the modal bin is $M2.2$, so the reported cut-off is $M_c = M2.4$, above the $M2.0$ at which this synthetic catalogue was constructed complete, and 41.8% of the events lie above it. The estimate is exposed as an explicit, user-adjustable cut-off rather than applied silently, so that the completeness assumption behind any subsequent b -value is visible at the point of use.

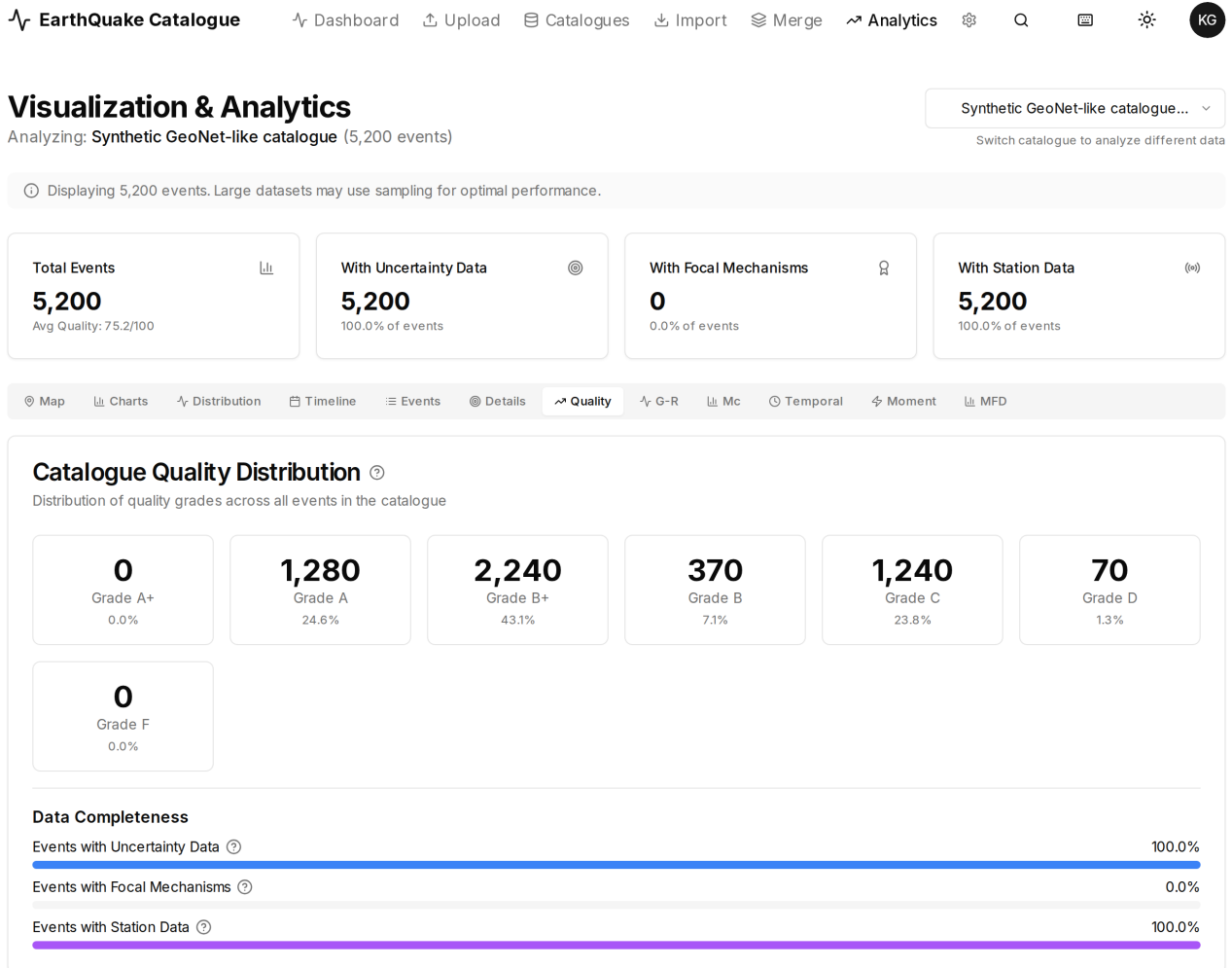


Figure S3: Catalogue-level quality summary for the synthetic GeoNet-like catalogue. The grade distribution and mean score are the five-dimension per-event index of equation 1 of the main paper, which scores location, network geometry, solution, magnitude, and evaluation status; every event is scored on import and mapped to an A⁺–F grade. The field-completeness bars are a separate catalogue-level metric that reports the fraction of events populating each optional field; they are not inputs to equation 1, and the absence of focal mechanisms in this synthetic catalogue therefore lowers the completeness bars without affecting any event’s score. No event reaches grade A⁺ because the synthetic events carry no station, phase, or uncertainty metadata, and equation 1 treats a missing field as its worst reportable value.